

SAGAR CHAUDHARY

Performance Engineering Lead | Site Reliability Engineer | AI/LLM Performance Engineer

Bangalore / Kolkata, India | +91 98046 44867 | sagar98chaudhary19@gmail.com

[LinkedIn](#) | [Portfolio](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Performance Engineering Lead with 5.6 years of hands-on experience in enterprise performance testing, Site Reliability Engineering (SRE), and AI/LLM observability across insurance, banking, and retail domains. Specialized in designing scalable cloud-native architectures, optimizing end-to-end latency, and ensuring SLA/SLO compliance. Pioneered LLM chatbot performance engineering using LangSmith and WebSocket protocol, tracking TTFT, token throughput, and per-session cost estimation. Expert in Apache JMeter, LoadRunner, NeoLoad, Dynatrace, Datadog, AWS, Azure, and Kubernetes. Proven delivery record: 40% database query optimization, 35% JVM heap improvement, 25% login response time reduction, and 71% LLM token cost reduction across mission-critical production systems.

CORE COMPETENCIES & TECHNICAL SKILLS

Performance Testing: Load, Stress, Spike, Volume, Scalability, Endurance, Soak, Regression, Baseline Testing, Capacity Planning, LLM/Chatbot Performance Testing

SRE & Reliability: SLI/SLO/SLA Design, Error Budget Management, Incident Management, RCA, High Availability, Fault Tolerance, Disaster Recovery, Chaos Engineering, MTTR/MTBF Optimization

AI/LLM Observability: LangSmith, TTFT, Token Throughput, Input/Output Token Analysis, Prompt Compression, Cost Optimization, WebSocket Performance, RAG Performance Testing, GPT-4o / Claude / Gemini Benchmarking

Performance Test Tools: Apache JMeter (Advanced), LoadRunner, NeoLoad, Gatling, OctoPerf, BlazeMeter, K6, Swagger/OpenAPI, Postman, JMeter MCP Server

APM & Monitoring: Dynatrace, Datadog, AppDynamics, New Relic, Grafana, Kibana, Prometheus, ELK Stack, AWS CloudWatch, Azure Application Insights, LangSmith

Cloud & DevOps: AWS (EC2, S3, EKS, CloudWatch, Auto Scaling, IAM, Lambda), Azure (AKS, Application Insights, Azure DevOps), Kubernetes, Docker, Jenkins, GitLab CI/CD, GitHub Actions, Terraform

Scripting & Programming: Java, Python, Groovy, JSR223, Shell Scripting, PowerShell, BeanShell, SQL

Databases: Oracle, SQL Server, MySQL, PostgreSQL, MongoDB — Query Optimization, Index Tuning, Execution Plan Analysis

Leadership: Team Leadership, Stakeholder Management, Agile/Scrum, Shift-Left Testing, Mentoring, Technical Presentations

PROFESSIONAL EXPERIENCE

Senior Associate – Performance Engineering | PwC India

Mar 2026 – Present

- Lead performance engineering strategy for enterprise AI-powered and cloud-native client engagements, improving user experience and system reliability.
- Pioneered LLM chatbot performance testing framework using LangSmith and WebSocket protocol to simulate real-time conversational load under 200+ concurrent users.
- Defined and tracked key LLM observability metrics: TTFT, total token latency, average tokens per request, input/output token counts, and per-session cost estimation.
- Reduced P99 latency from 7s to 1.8s and achieved 71% token cost reduction through prompt compression (850 to 120 tokens) and max_tokens capping at 250.
- Configured dual-platform observability integrating LangSmith for trace-level LLM analysis and Datadog for infrastructure monitoring, enabling end-to-end performance visibility.
- Benchmarked 5 LLM models (GPT-4o, GPT-4o-mini, Claude 3 Sonnet, Claude 3 Haiku, Gemini 1.5 Flash) for insurance chatbot; recommended Claude 3 Haiku based on lowest TTFT (0.7s), cost, and stability.
- Developed SLI/SLO/SLA monitoring dashboards using Dynatrace and Datadog, ensuring 99.9% availability compliance across production services.
- Integrated performance testing into Agile CI/CD pipelines using Jenkins and GitHub Actions, enabling shift-left performance engineering.

Performance Test Engineer | Capgemini

2025 – 2026

- Designed end-to-end performance testing strategy for RLI Insurance Portal covering load, stress, spike, and endurance scenarios, ensuring high availability under peak production load.
- Analyzed APM metrics including CPU utilization, JVM garbage collection, API response latency, and database lock contention using Dynatrace and Datadog.

- Reduced login response time by 25% through targeted JVM tuning, thread pool optimization, and microservices-level refactoring, directly improving end-user experience.
- Leveraged AI-assisted tooling with Swagger/OpenAPI specifications to accelerate JMeter script generation and critical user journey identification, cutting script development time by 40%.
- Validated AI-generated test artifacts manually to ensure accuracy and compliance with performance KPIs and business SLAs.

Performance Test Engineer | Tata Consultancy Services (TCS)

2022 – 2025

- Executed high-volume performance tests for a US Banking client, simulating 50,000+ concurrent users using JMeter and LoadRunner on distributed architecture.
- Implemented distributed JMeter execution on AWS EC2 with Auto Scaling for elastic load generation, reducing test infrastructure costs by 30%.
- Architected microservice latency and infrastructure health dashboards integrating AWS CloudWatch, Datadog, and AppDynamics for real-time incident detection and RCA.
- Delivered 40% improvement in database query performance through SQL optimization, composite index tuning, and execution plan analysis on Oracle.
- Led regulatory performance testing initiatives ensuring SLA compliance and supporting audit-ready documentation for banking performance thresholds.
- Mentored 4 junior engineers on JMeter scripting, correlation, and APM analysis best practices.

Performance & Automation Engineer | Wipro

2020 – 2022

- Performed scalability and load testing for OmniStore retail microservices (15+ services) on Kubernetes EKS and AKS using JMeter and Dynatrace, ensuring reliability under peak traffic.
- Identified and resolved GC pauses, memory leaks, and thread contention through deep JVM profiling, reducing production incidents by 45%.
- Achieved 35% JVM heap optimization and validated Kubernetes Horizontal Pod Autoscaler (HPA) behavior under sustained and spike load conditions.
- Built automated performance regression test suites integrated with Jenkins and GitLab CI/CD pipelines to enable shift-left performance engineering.
- Configured Kafka consumer lag monitoring and Grafana dashboards with Dynatrace for real-time observability and proactive incident detection.

KEY PROJECTS

LLM Chatbot Performance Engineering – PwC AI Observability Initiative

Tech Stack: JMeter • WebSocket • LangSmith • Datadog • Dynatrace • Python • Azure AKS • Groovy JSR223

- Designed performance testing framework for enterprise LLM chatbot using WebSocket protocol to simulate real-time conversational load at scale.
- Tracked TTFT, total response latency, tokens per request, input/output tokens, and per-session cost via LangSmith and custom Datadog histograms.
- Outcome: Improved TTFT from 7s to 1.8s (74% reduction) and cut per-session LLM cost by 71% via prompt compression.

RLI Insurance Portal – End-to-End Performance Engineering

Tech Stack: JMeter • OctoPerf • Dynatrace • Datadog • Azure AKS • SQL Server

- Designed testing strategy covering load, stress, spike, and endurance scenarios with full CI/CD integration via Jenkins.
- Identified bottlenecks across API gateway, microservices, and database tiers under 5,000 concurrent user load.
- Outcome: Achieved 25% login response time improvement through JVM and microservice-level tuning.

US Banking – High-Volume & Regulatory Performance Testing

Tech Stack: JMeter • LoadRunner • AWS CloudWatch • Datadog • AppDynamics • Oracle

- Simulated 50,000+ concurrent users for core banking workflows under peak load on distributed architecture.
- Implemented distributed load generation on AWS EC2 with auto-scaling for elastic, cost-efficient test execution.
- Outcome: Delivered 40% database performance improvement through SQL query optimization and index restructuring.

OmniStore Retail Microservices – Kubernetes Scalability

Tech Stack: JMeter • Dynatrace • Datadog • Kubernetes EKS/AKS • Kafka

- Validated HPA behavior under sustained and spike load for 15+ microservices on Kubernetes.
- Identified critical JVM issues (GC pauses, memory leaks, thread contention) via Dynatrace deep-dive profiling.
- Outcome: Achieved 35% JVM optimization and established performance baselines for regression benchmarking.

OPEN SOURCE PERFORMANCE TOOLS (GITHUB)

LLM Performance Observability Dashboard

Tech Stack: *Python • Streamlit • Plotly • Pandas*

- Built interactive dashboard benchmarking 5 LLM models (GPT-4o, Claude, Gemini) across TTFT, P99 latency, token throughput, and cost per session.
- Includes configurable SLO thresholds, auto model recommendation engine, and time-series latency analysis.

Live: llm-performance-dashboard.streamlit.app

JMeter AI Script Generator

Tech Stack: *Python • Streamlit*

- Generates production-ready JMeter JMX scripts from Swagger/OpenAPI specs — includes Thread Groups, Assertions, CSV DataSet, Datadog Backend Listener, Header Manager.
- Supports Bearer Token, API Key auth; configurable ramp-up, duration, think time, and response timeouts.

Live: jmeter-ai-generator.streamlit.app

Performance Test Report Generator

Tech Stack: *Python • Streamlit • Plotly • Pandas*

- Upload JMeter JTL files and instantly generate professional HTML reports with P50/P95/P99 charts, SLO PASS/FAIL validation, error breakdown, and throughput analysis.
- Exports standalone HTML report and CSV summary stats — ready to share with stakeholders.

Live: perf-report-generator.streamlit.app

RELIABILITY & OBSERVABILITY HIGHLIGHTS

- SLI/SLO/SLA Design & Monitoring: defined error budgets and performance thresholds across cloud-native services.
- Incident Management & RCA: led post-incident reviews to eliminate recurring performance degradations and reduce MTTR.
- Distributed Systems Observability: integrated multi-tool monitoring stacks (Dynatrace, Datadog, LangSmith, Grafana, Prometheus) for unified visibility.
- High Availability & Resilience: validated failover, autoscaling, and chaos scenarios under spike and endurance conditions.

EDUCATION & CERTIFICATIONS

B.Tech – Electronics & Communication Engineering | Graduated: 2020

Certifications:

- ISTQB Certified Tester Foundation Level (CTFL)
- Microsoft Azure Fundamentals – AZ-900
- Apache JMeter Performance Testing Certification
- Deloitte & Accenture Virtual Job Simulations

PORTFOLIO & LINKS

Portfolio: sagar-portfolio-new.vercel.app | **GitHub:** github.com/sagar9804644867